Nigerian Journal of Parasitology
ISSN 1117 4145 Volume 43[1] March, 2022

Sero-Prevalence of Syphilis and Assessment of Haemoparasites among Prospective Blood Donors at a Tertiary Hospital in Calabar, Southern Nigeria.

***Okafor, I. M.[1], Akpan S. S.[2] and Henshaw, A. E.[1]**
[1]Department of Haematology and Blood Transfusion Science, Faculty of Medical Laboratory Science, College of Medical Sciences, University of Calabar, Cross River State, Nigeria.
[2]Department of Medical Parasitology and Entomology, Faculty of Medical Laboratory Science, College of Medical Sciences, University of Calabar, Cross River State, Nigeria.
email: okaforify12@gmail.com; okaforify@unical.edu.ng

Abstract
The prevalence of haemoparasites among blood donors in the studied area has not been previously documented. This study determined the prevalence of syphilis and haemoparasitic infections among blood donors in Calabar, Cross River State, Nigeria. This was a hospital-based cross-sectional study involving 200 consenting blood donors. Samples were processed using the Venereal Disease Research Laboratory (VDRL) screening technique for the detection of *Treponema pallidum* antibodies. Screening for microfilaria was conducted using Knott's concentration technique. Thick and thin blood films stained with 3% Giemsa solution were examined to confirm the presence of malaria parasites. All the 200 blood donors harboured trophozoites of *Plasmodium falciparum*, with low parasite counts per millilitre of blood. One hundred and thirteen (56.5%) blood donors had microfilaria *Loa*. 21(10.5%) blood donors were positive for antibodies homologous to *T. pallidum* antigens. Mixed infections of *P. falciparum* + Microfilaria; *P. falciparum* + Microfilaria + *T. pallidum* and Microfilaria + *T. pallidum* occurred at the prevalence rates of 50%, 16.5% and 4%, respectively. The prevalence of haemoparasites was higher among male blood donors than in their female counterparts. The most prevalent infection based on the frequency of blood donation was the malaria parasite (45.5%). Among the first-time blood donors, the most prevalent infections were malaria (54.5%) and *Loa loa* (24.5%). This study has shown a significantly ($p < 0.05$) high prevalence of malaria parasite and *Loa loa* infection in the studied group, it also showed malaria parasite and *Loa loa* as the major haemoparasites found among blood donors in the health institution studied.

Keywords: Haemoparasite, Microfilaria, Syphilis and Transfusion.

Introduction

Blood is a fluid connective tissue that circulates throughout the body through the arteries and veins providing a vehicle by which an immense variety of different substances are transported between the various organs and tissues [1]. Blood donation is the giving of blood by a healthy person for storage in a blood bank and subsequent transfusion into another person [1]. The blood for transfusion is screened routinely for transfusion transmissible infections such as human immunodeficiency virus (HIV), hepatitis B virus (HBV), hepatitis C virus (HCV) and *Treponema pallidum* (agent of syphilis), but

The foregoing study on pages 41-46 was submitted in February 2022.
https://dx.doi.org/10.4314/njpar.v43i1.7
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screening of parasitic infections such as *Trypanosoma cruzi* (agent of Chagas' disease) and microfilaria (agent of filariasis) are not routinely done in most blood transfusion centres. Other pathogens which can also be transmitted in the blood include *Brucella* species, *Cytomegalovirus, Epstein bar virus* and *Babesia* species [2,3]. The recipients of unscreened blood infected with any of these pathogens are exposed to these infections which in most cases can cause disease or even death of the recipients. Therefore, screening of blood before donation and transfusion is especially important in blood transfusion practice. Though transfusion therapy is a form of treatment based on the use of blood and its products from humans, which helps to save lives, the practice can also be a dreadful vehicle for the transmission of some agents of infectious diseases [4,5]. Previous researchers had reported the prevalence of some of the haemoparasites (malaria parasite and microfilariae) in this locality [6,7,8]. Therefore, this study is expected to provide information on the prevalence of syphilis and haemoparasitic infections among prospective blood donors.

Materials and Methods

This is a cross-sectional study conducted among subjects who came to donate blood at the donor clinic of the University of Calabar Teaching Hospital, Calabar, Southern Nigeria. Using systematic random sampling, 200 subjects were selected out of the number of prospective donors that came. Their ages ranged between 18 and 65 years, the donors were comprised of one hundred and forty male donors and sixty female donors. People who had skin rashes, purpura, and those whose red cell-specific gravity was less than 1.055, packed cell volume less than 40 per cent for male donors and less than 38 per cent for female donors or people who had undergone surgery or accident within the last four months and also pregnant women and nursing mothers were excluded. Inclusion criteria were those with red blood cell-specific gravity of more than 1.055, packed cell volume of more than 40 per cent for males and 38 per cent for females and those who had not undergone surgery. Blood samples were collected through venepuncture from the most prominent veins. The samples were placed in plain bottles to obtain serum for the VDRL test for syphilis and also immediately put on a slide to prepare thick and thin blood film for the detection of haemoparasites.

Ethical approval and informed consent

Ethical clearance (CRS/MH/CGS/018/vol.1/21) was sought and obtained from the Cross River State Committee on Ethics after which written permission for the study was given by the department and patients' informed consent were obtained before the study. Structured questionnaires were administered with the aid of health workers to the enrolled subjects to obtain their bio-data and other relevant information.

Detection of malaria parasites

Thick and thin blood films were made on clean, grease-free slides. They were air-dried. The thin films were fixed in absolute methyl alcohol briefly and then stained with 3% Giemsa solution for 30 minutes. The stain was rinsed afterwards with water. The slides were air-dried and then examined microscopically using X100 objective lens for the detection of malaria parasites.

Knott's concentration technique (1939) [9] for the demonstration of microfilaria.

Nine millilitres of 1% formalin solution were added to 1 millilitre of venous blood sample in a centrifuge tube and centrifuged at 3000 revolutions per minute for 5 minutes. The supernatant was poured off and the tube was tapped to re-suspend the deposit. The entire deposit was used to make blood smears on clean, grease-free slides and left to stand on the bench until the smears were air-dried completely. The blood smears for detection of microfilaria were stained with 3% Giemsa solution for 30 minutes. The slides were air-dried. For microscopy, X10 objective lens was used for scanning through. Microfilariae were identified microscopically using X100 objective lens after applying immersion oil.

Screening test for syphilis

A VDRL test kit (VIKIA® syphilis kit by bioMérieux SA, France) was used. The kit's manufacturing date was February 2017 and the expiring date was February 2021. The strip was used according to manufacturing instruction by dipping into the serum, brought out and left for 10 minutes. The appearance of two-coloured lines indicated a positive result while one line indicated a negative result.

Statistical Analysis

The results were presented in Tables and data were analysed using Chi-Square.

Results

Table 1 shows the prevalence of syphilis and common haemoparasites among the prospective blood donors in the hospital. *Plasmodium falciparum* infection was the most prevalent (100%) followed by microfilaria *Loa loa* (56.5%) and syphilis (10.5%). The table also shows that the prevalence of malaria parasite infection was significantly higher than that of microfilaria and syphilis infection among the blood donors ($p< 0.005$). Table 2 is on the number of prospective blood donors with single infection, mixed infections, or no infection. The prevalence of prospective blood donors with a single infection was (100%) while those with mixed infection was (60.5%). The prevalence of infected subjects was (100%) and there was no negative sample. Table 3 shows the prevalence of syphilis and common haemoparasitic infections among prospective male and female blood donors. The result showed that male (69.5%) and female (30.5%) malaria-infected subjects were most prevalent followed by microfilaria male (45.5%) and female (11.0%) infected subjects and syphilis infected males (7.0%) and females (3.5%) subjects. The table also shows that malaria, microfilaria, and syphilis infections were significantly ($p< 0.05$) higher in male blood donors when compared to female blood donors. Table 4 shows the prevalence of syphilis and common haemoparasitic infections based on the frequency of blood donation. The result showed that first-time blood donors (92.0%) were more infected significantly ($p < 0.05$) than frequent blood donors (75.0%).

Discussion

The findings of the present study showed that malaria caused by *P. falciparum* was the most prevalent blood parasitic infection among blood donors, followed by filariasis (*Loa loa*) and syphilis infection (*Treponema pallidum*) in Calabar, Cross River State. All the prospective blood donors were positive for the malaria parasite. However, they did not show any sign of clinical malaria.

This supports the earlier study conducted in Ilorin, Nigeria [8]. Out of the 308 donors they screened, the parasites detected were mostly *Plasmodium falciparum* (85.7%) while the prevalence of *T. pallidum* was 3.6%. A high prevalence rate of malaria caused by *Plasmodium falciparum* (23.4%) was also reported in Nnewi, Nigeria [7]. The high prevalence rate of malaria can be attributed to poor sanitary and environmental conditions, ignorance of the dangers of the infection, low socioeconomic status as well as the absence of preventive measures. Furthermore, malaria parasite infection is also endemic in Calabar, Cross River State [4].

This study has shown a high prevalence rate of *Loa loa* (56.5%) in the studied group. A previous study reported up to 18% prevalence of *L. loa* from certain parts of Cross River State [10]. Similar reports confirmed *L. loa* and *M. perstants* as part of the 14 parasitic infections recorded among nomadic Fulani in South-eastern Nigeria [10,11]. In a related study [12] conducted in Myanmar and Thailand, out of 400 donors screened, the prevalence rate of *Loa loa* was 2.4% and 1.3%, respectively. These findings are low, compared to the result (56.5%) found in this study. The present study has confirmed the continued prevalence of microfilaria, *Loa-loa* in Calabar, Nigeria. *Chrysops* are important vectors of these filarial worms called *Loa loa*, studies have shown a high prevalence (70.5%) of this vector in Cross River State [13, 14]. This may account for the high prevalence of *Loa loa* recorded in this study.

This study has also shown that malaria caused by *Plasmodium falciparum* was the most prevalent among those who had a single infection. Among those who had mixed infections were malaria and microfilaria (*Loa loa*) (50%) followed by those with malaria, microfilaria (*Loa loa*) and *T. pallidum* (16.5%) then *Treponema pallidum* and microfilaria (4.0%). This supports the earlier research conducted in Calabar, Cross River State [14]. The findings of this research have shown that the prevalence rates of malaria parasite infection among male and female donors were 69.5% and 36% respectively followed by microfilaria (*Loa loa*) (45.5%) and (11.0%) respectively and then *T. pallidum* (7.0%) and (3.5%). This agrees with research conducted among donors in a rural area of Gambia where the prevalent rate of microfilaria (*Loa loa*) for males and females were (48.0%) and (50.0%) respectively [16].

This research has also shown a high prevalence rate of malaria (54.5%) followed by microfilaria (*Loa loa*) (82.0%) and *Treponema pallidum* (5.5%) among first-time donors who participated in the study. Among the frequent donors, malaria (45.5%) was most prevalent ($p < 0.05$) followed by microfilaria (*Loa loa*) (24.5%) and *T. pallidum* (5.0%). These findings agree with previous research done in Calabar, Cross River which showed that among the first-time blood donors who participated in this study,

malaria (93.7%) and *Loa loa* (29.5%) were the most prevalent while among the frequent donors, malaria (39.2%) and microfilaria (*Loa loa*) (19.0%) were the most prevalent [17].

Routinely, prospective blood donors who had *T-pallidum* are not bled to avoid transmission to the recipient, but this is not the case in parasites like microfilaria (*Loa loa*) and malaria parasites. Studies have shown that when blood is stored at a temperature of 4°C ± 2°C, the parasite (*Loa loa*) becomes inactive within a few hours and will not be harmful to the recipients [18]. But in cases of emergency, where the blood is transfused without storage, the recipient is at risk of being infected with malaria parasite or microfilaria (*Loa loa*). In cases where a patient is mistakenly transfused with malaria parasite positive blood, the patient can be given a curative regimen of anti-malarial drug, especially when the individual falls within the malaria vulnerable group as it is practised in this part of the world, but this is not the case with *Loa loa*.

In conclusion, this study has shown a high prevalence of microfilaria (*Loa loa*) among the blood donors in a tertiary hospital in Calabar, Nigeria. The study also shows that syphilis infection is still a cause for concern among blood donors in this locality. We, therefore, recommend a continuous creation of awareness for effective control and elimination of these infections.

Table 1: The Prevalence of syphilis and common haemoparasitic infections among prospective blood donors in UCTH.

TEST	No. (%) of positive sample	No. (%) of negative sample
VDRL	21 (10.5)	179 (89.5)
MP	200 (100)	0 (0.0)
MF	113 (56.5)	87 (43.5)
P-Value	P < 0.05	

Key: VDRL- Venereal Disease Research Laboratory, MP- Malaria Parasite, MF- Microfilaria

Table 2: Prospective blood donors with single and mixed infections

Parameter	No. (%) mixed/single infection
MP	79 (100)
MP + MF	100 (50)
MF + VDRL	8 (4.0)
MP + MF + VDRL	13 (16.5)
P-Value	P < 0.05

Key: VDRL- Venereal Disease Research Laboratory, MP- Malaria Parasite, MF- Microfilaria

Table 3: The Prevalence of syphilis and common haemoparasitic infections among prospective male and female blood donors.

Test	No. (%) of male positive donors (n=140)	No. (%) of female positive donors (n=60)
VDRL	14 (7.0)	7 (3.5)
MP	140 (69.5)	60 (30.0)
MF	91 (45.5)	22 (11.0)
P-Value	P < 0.05	

Key: VDRL- Venereal Disease Research Laboratory, MP- Malaria Parasite, MF- Microfilaria

Table 4: The Prevalence of syphilis and common haemoparasitic infections based on the frequency of blood donation

Test	No. (%) of positive first-time donors (n=87)	No. (%) of positive frequent donors (n=112)
VDRL	11 (5.5)	10 (5.0)
MP	109 (54.5)	91 (45.5)
MF	64 (32.0)	49 (24.5)
Total	184 (92.0)	150 (75.0)
P-Value	P < 0.05	

Key: VDRL- Venereal Disease Research Laboratory, MP- Malaria Parasite, MF- Microfilaria

Acknowledgements
The authors are grateful to all the Prospective blood donors that participated in this study and also to the staff of the Hospital for their cooperation and contribution during the study.

Funding
The authors funded this research

Disclosure of Conflict of Interest
The authors declare no conflict of interest

Author's contributions
Okafor, I. M conceived the study, performed sample analysis, performed literature search, supervised the work, wrote, and edited the manuscript draft. Akpan, S. S. also wrote and edited the manuscript draft, Henshaw, A. E. performed the literature search, sample, and statistical analysis. All authors read and approved the final manuscript.

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Citation
Okafor, I. M., Akpan, S. S. and Henshaw, A. E. (2022). Sero-Prevalence of Syphilis and Assessment of Haemoparasites among Prospective Blood Donors at a Tertiary Hospital in Calabar, Southern Nigeria. *Nigerian Journal of Parasitology*, 43(1) 41-46

https://dx.doi.org/10.4314/njpar.v43i1.7

Nigerian Journal of Parasitology
ISSN 1117 4145 Volume 43[1] March, 2022